

Modeling Demand Pressure Curves in Liquidity Bootstrapping Pools

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Abstract

Liquidity Bootstrapping Pools (LBPs) have become a widely adopted mechanism for fair token distribution in decentralized finance. Despite their popularity, systematic methods for characterizing demand dynamics during LBPs remain limited. This paper proposes a data-driven framework to model demand pressure curves using historical on-chain data from the Balancer ecosystem. By clustering LBPs according to a composite success metric and anchoring representative behavior using modal statistics, we derive characteristic demand profiles for high-, medium-, and low-success launches. The resulting curves provide a practical foundation for simulation, forecasting, and improved LBP parameter design.

1 Introduction

Liquidity Bootstrapping Pools (LBPs) are designed to enable fair price discovery while mitigating early speculative dominance. However, LBPs exhibit heterogeneous outcomes, ranging from highly successful launches with sustained demand to rapid collapses driven by short-term speculative activity. This work introduces a quantitative methodology to model demand pressure curves across different LBP outcomes, grounding simulations in empirical market behavior rather than heuristic assumptions.

2 Data Collection

The empirical foundation of this study consists of 961 historical LBPs extracted from the Balancer V1 and V2 protocols across Ethereum Mainnet, Polygon, and Arbitrum. Trade-level data were retrieved using Dune Analytics queries, capturing the full lifecycle of each pool, including swap timestamps, traded amounts, and USD-normalized volumes.

The analysis builds upon prior work on LBP success classification, reusing established metrics while extending them toward curve modeling. A dedicated query was used to extract all swaps for each pool, enabling the reconstruction of price trajectories and demand dynamics over time.

```
1 SELECT
2     block_time ,
3     token_pair ,
4     token_bought_symbol ,
5     token_sold_symbol ,
6     token_bought_amount ,
7     token_sold_amount ,
8     amount_usd ,
9     pool_symbol ,
10    pool_type ,
```

```

11     taker AS swapper ,
12     tx_hash
13 FROM balancer.trades
14 WHERE project_contract_address = {{pool_address}}
15      OR pool_id = {{pool_address}}
16 ORDER BY block_time DESC;

```

Listing 1: Query used to extract LBP swap data

Table 1: LBP success metrics used for clustering

Metric	Source	Description
pool_address	Raw	Foreign key (links to the pool registry)
volume_usd	Calculated	Total USD traded during the LBP
unique_buyers	Calculated	Number of unique buyers
price_retention	Calculated	Average price (last window) / average price (first window)
volatility_score	Calculated	Mean price / price standard deviation
dump_pressure	Calculated	Buy volume / sell volume
bot_tx_ratio	Calculated	Trades in the first window / total trades
final_success_score	Derived	Composite success score used for clustering

3 Success Metrics and Clustering

Each LBP is evaluated using a composite success score derived from multiple dimensions: price retention, volatility, buyer diversity, bot participation, and buy-sell imbalance. Pools are classified into three clusters:

- High success: success score > 0.8
- Medium success: $0.4 < \text{success score} \leq 0.8$
- Low success: success score ≤ 0.4

This clustering enables the comparison of structurally distinct demand behaviors under different market conditions.

4 Curve Modeling Methodology

4.1 Exploratory data analysis

From the histogram plots generated for each column in the dataset, it is possible to observe that the data do not follow a normal distribution. By definition, this invalidates the use of the mean as the anchoring metric, since the mean is highly sensitive to skewness and extreme values, and therefore fails to provide a representative measure of central tendency in non-Gaussian distributions.

Additionally, the data exhibit highly concentrated distributions around specific value ranges, which favors the selection of the mode as the anchoring metric. In this context, the mode represents the most frequently observed behavior within each cluster, making it an appropriate statistic for capturing the typical shape of the curves while remaining robust to outliers and distributional asymmetries.

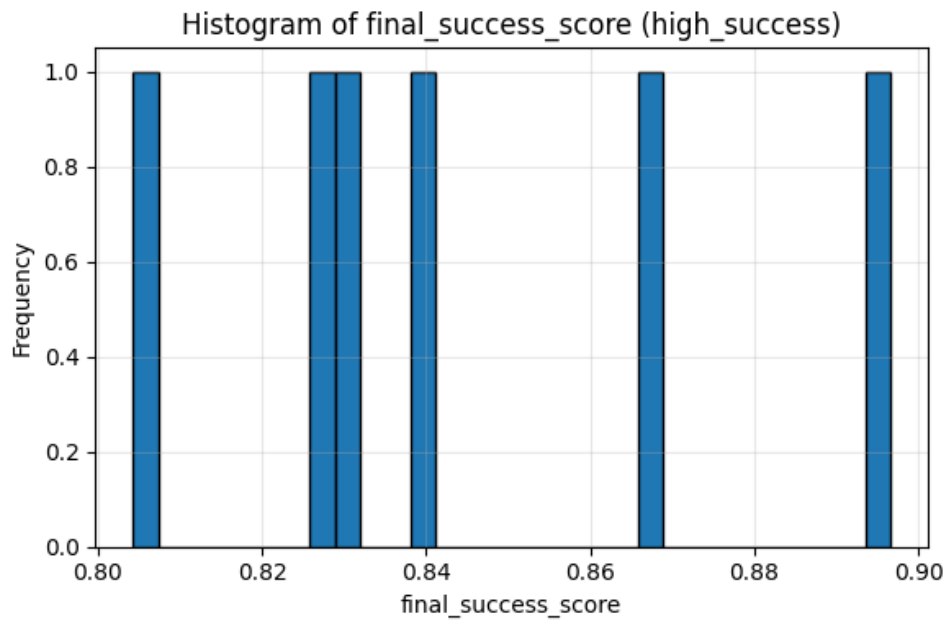


Figure 1: Success score histogram for the high-success cluster.

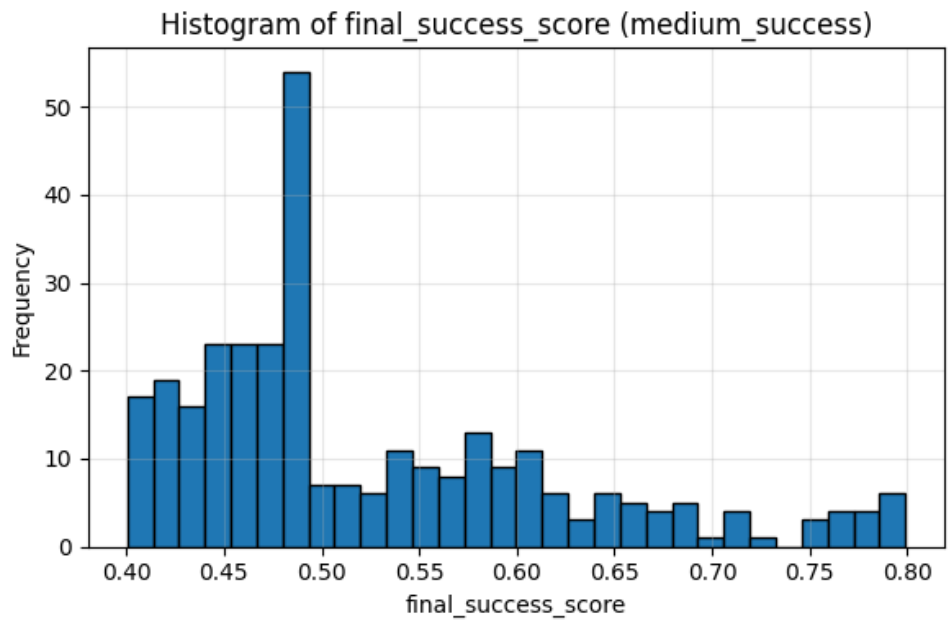


Figure 2: Success score histogram for the medium-success cluster.

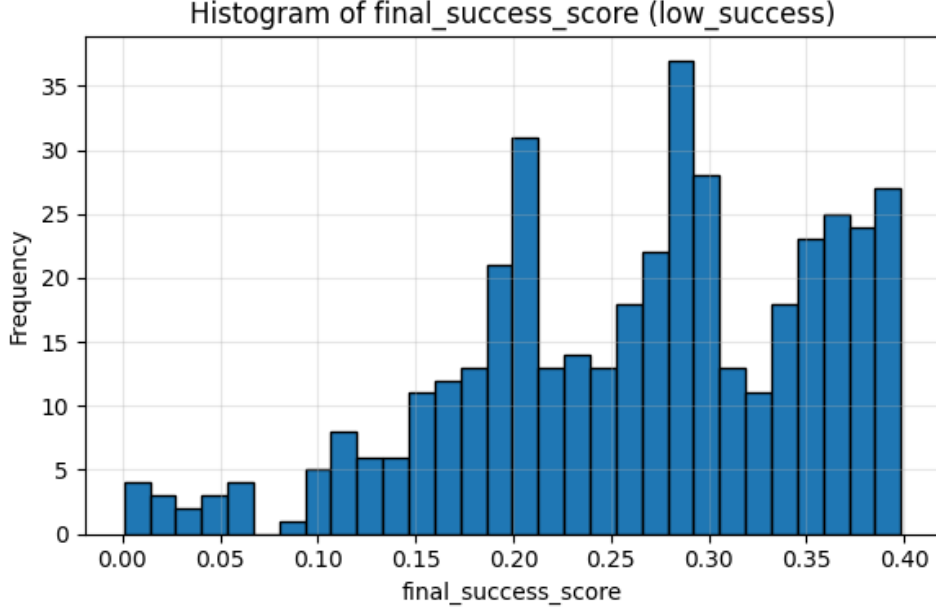


Figure 3: Success score histogram for the low-success cluster.

4.2 Anchoring statistic and representative pool selection

Exploratory analysis of the underlying distributions reveals strong skewness and non-Gaussian behavior, rendering mean-based aggregation unsuitable. Instead, the mode is selected as the anchoring statistic, as it captures the most frequently observed behavior within each cluster while remaining robust to outliers.

For each cluster, a representative pool is selected by minimizing the Euclidean distance between individual pool feature vectors and the cluster modal vector. This pool serves as a proxy for the typical demand curve of the cluster.

4.3 Selecting the reference pools

With the mode defined as the anchoring metric for the analysis, the next step is to construct the plots that validate the curve shape of each cluster. To determine which pool should be used as the reference for these plots, an approximation was performed using the Euclidean distance between each pool within the cluster and the vector of modal values computed for each column.

This approach allows us to identify the pool whose feature values most closely align with the modal values of the cluster, thereby serving as a representative instance. As a result, the selected pool effectively simulates the curve model that occurs most frequently within the dataset, providing a robust and interpretable representation of the cluster’s dominant behavior.

The results obtained for each of the three clusters are presented below. For each cluster, the selected pool corresponds to the instance whose feature vector exhibits the minimum Euclidean distance to the modal values of the cluster, thereby serving as the representative pool for the respective success category.

Table 2: High-success cluster representative pool metrics

Metric	Value
Pool address	0x5539a8dd01188f359993cff7f0bef5178164af9b
Volatility score	0.5407
Price retention	44.7772
Unique buyers	723
Bot transaction ratio	0.0012
Volume (USD)	1,993,599.82
Dump pressure	0.7794
Final success score	0.8043

Table 3: Medium-success cluster representative pool metrics

Metric	Value
Pool address	0x0a336d1a947dc92855a228f653f71f6905729b27
Volatility score	0.4483
Price retention	0.7086
Unique buyers	257
Bot transaction ratio	0.0014
Volume (USD)	2,035,253.67
Dump pressure	0.6019
Final success score	0.3597

Table 4: Low-success cluster representative pool metrics

Metric	Value
Pool address	0x5db3d8a689631d17f22dbd9ab16c51f561d20ffb
Volatility score	0.6230
Price retention	2.2769
Unique buyers	473
Bot transaction ratio	0.0049
Volume (USD)	2,181,658.67
Dump pressure	0.2320
Final success score	0.6440

5 Results

For each representative pool, we analyze:

1. Token price evolution over time,
2. Cumulative buy pressure,
3. Accumulated volume and individual swaps over time.

5.1 High-Success Pools

High-success pools exhibit sustained cumulative demand, controlled volatility, and price stabilization after the initial discovery phase.

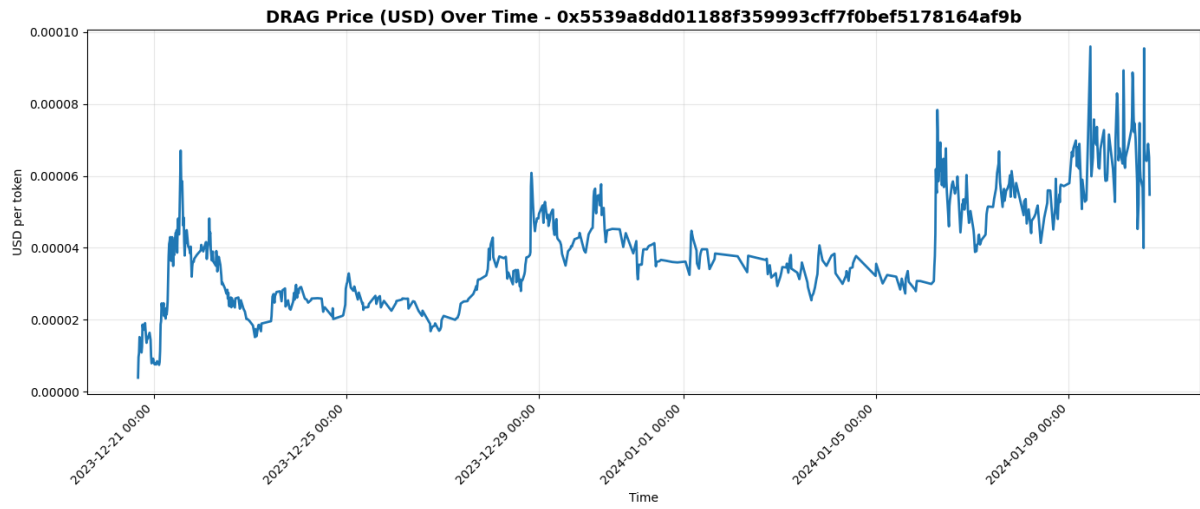


Figure 4: Token price over time for a high-success LBP.

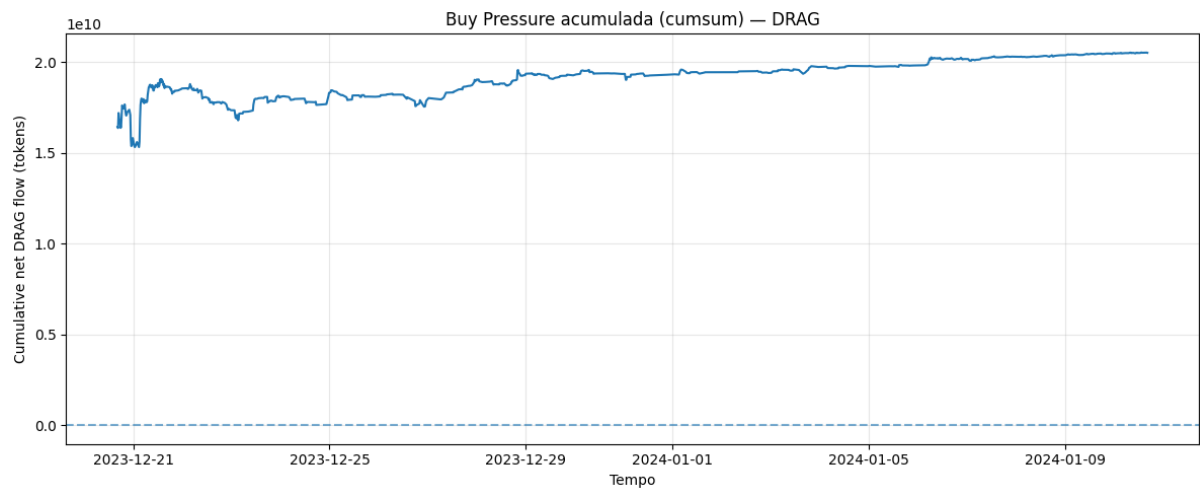


Figure 5: Cumulative buy pressure over time for a high-success LBP.

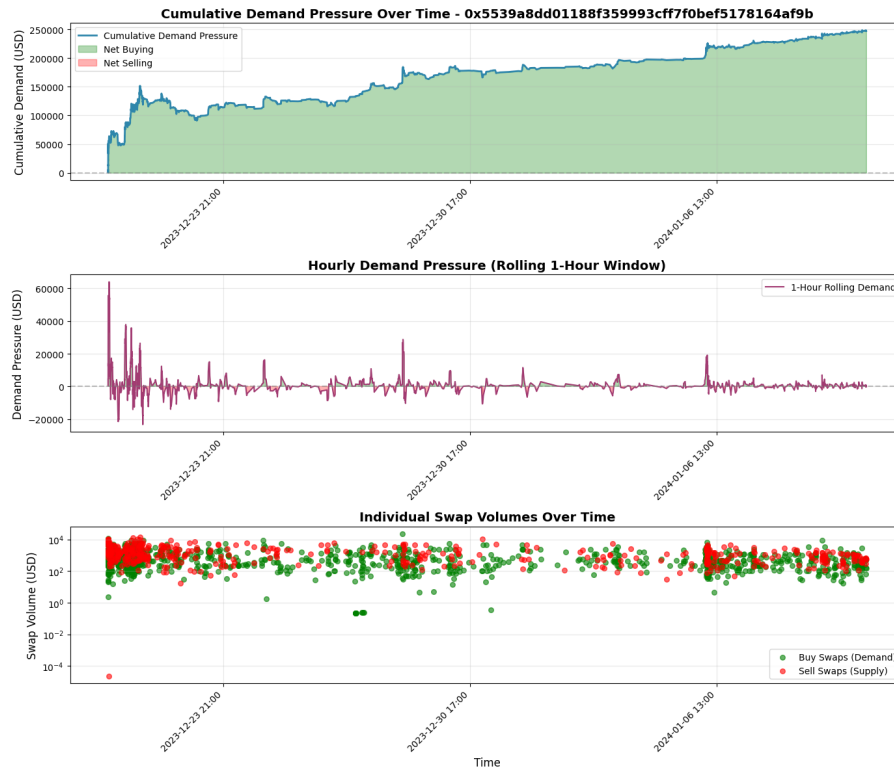


Figure 6: Accumulated volume and individual swaps over time for a high-success LBP.

5.2 Medium-Success Pools

Medium-success pools show early volatility followed by partial stabilization, indicating moderate conviction and intermittent sell pressure.

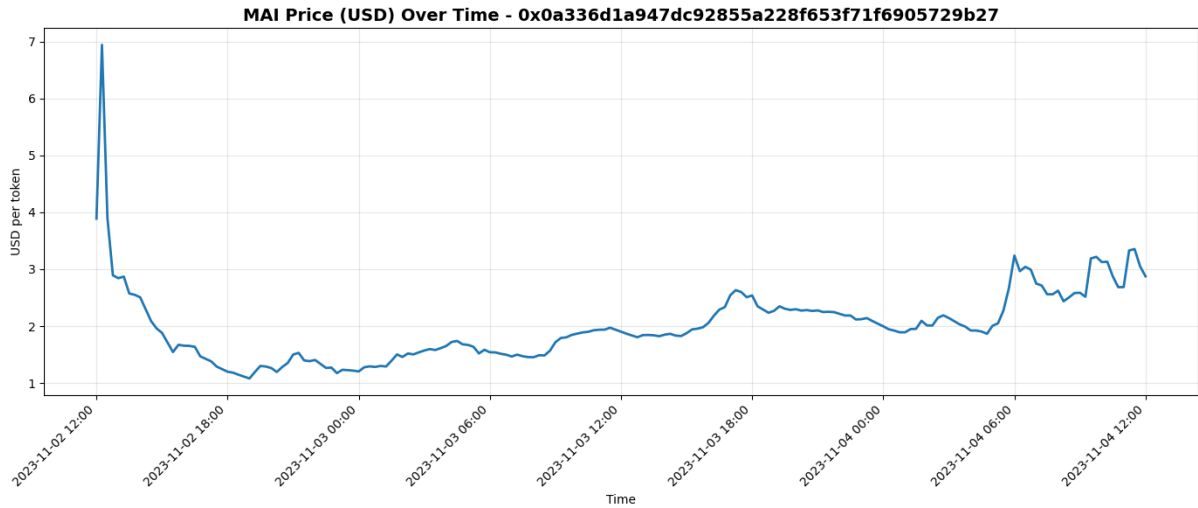


Figure 7: Token price over time for a medium-success LBP.

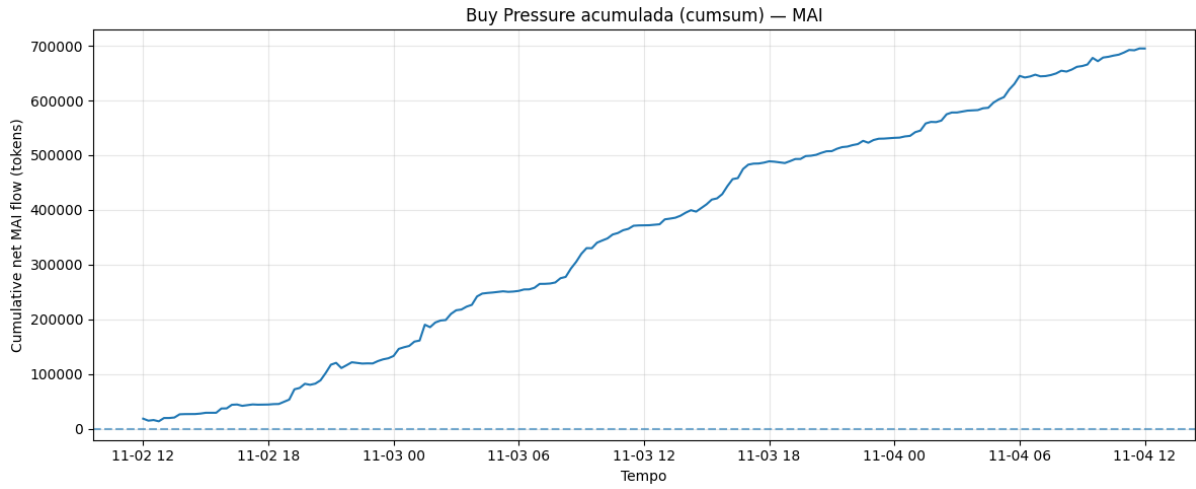


Figure 8: Cumulative buy pressure over time for a medium-success LBP.

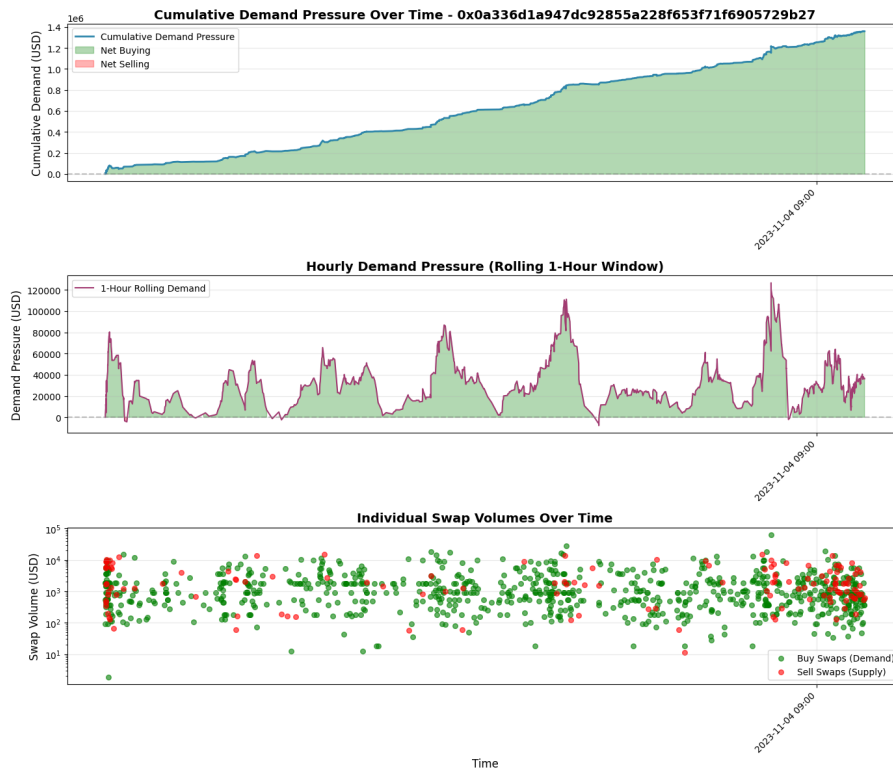


Figure 9: Accumulated volume and individual swaps over time for a medium-success LBP.

5.3 Low-Success Pools

Low-success pools are characterized by sharp initial demand spikes followed by rapid decay, consistent with speculative participation and weak long-term demand.

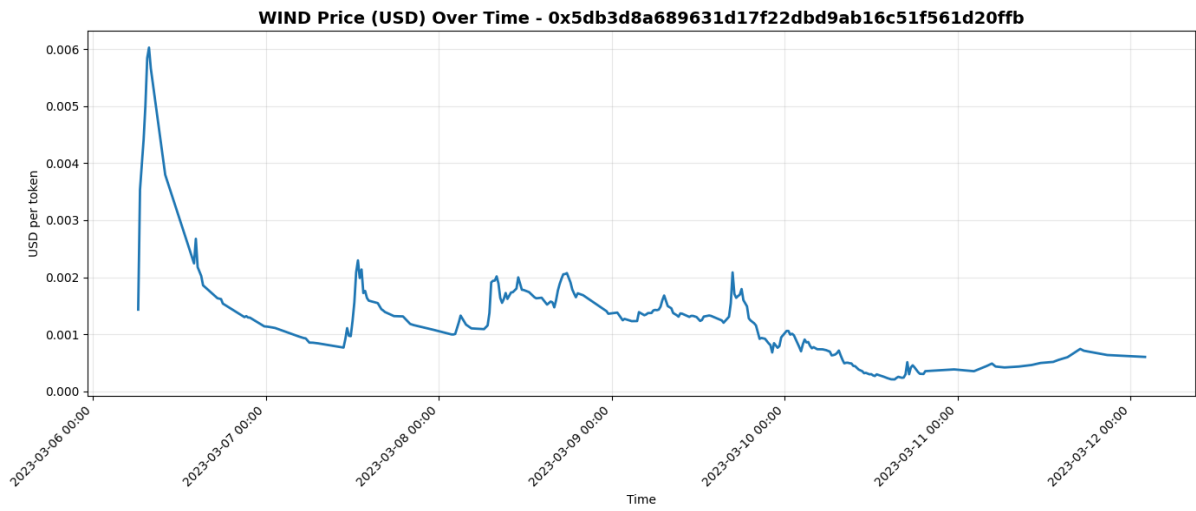


Figure 10: Token price over time for a low-success LBP.

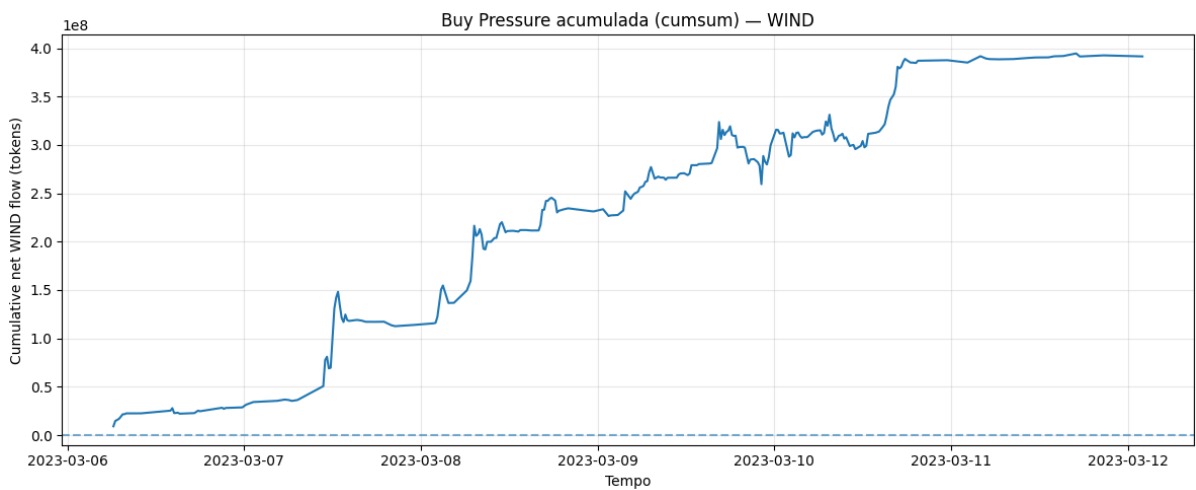


Figure 11: Cumulative buy pressure over time for a low-success LBP.

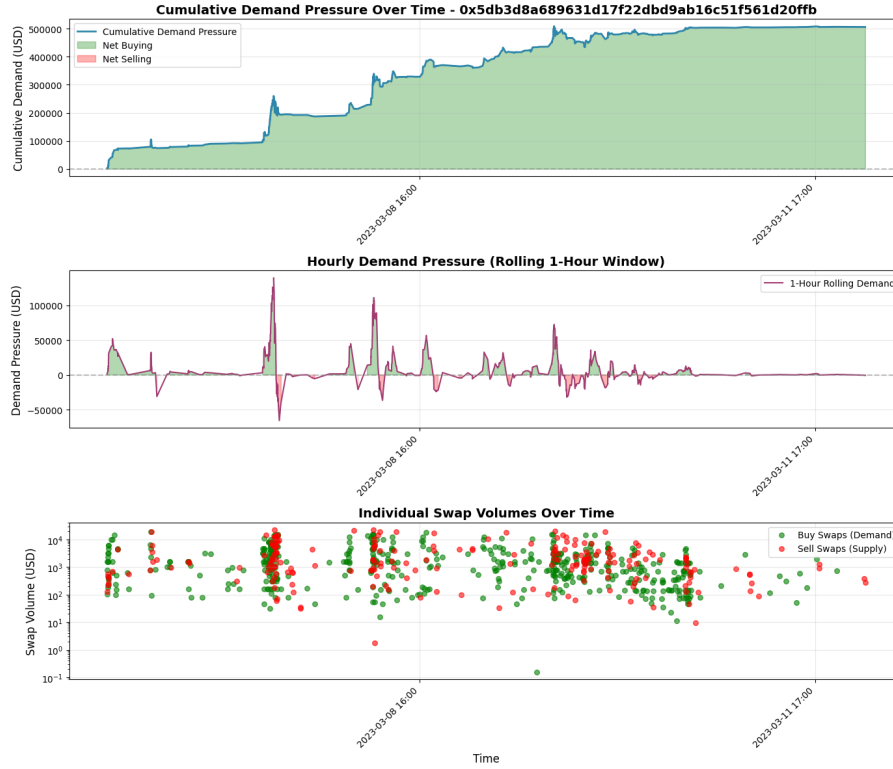


Figure 12: Accumulated volume and individual swaps over time for a low-success LBP.

6 Implications for LBP Design

The derived demand pressure curves provide actionable guidance for protocol designers and token teams. By aligning pool parameters and launch strategies with empirically observed demand profiles, LBPs can be designed to better match desired outcomes.

7 Conclusion

This study demonstrates that demand pressure during LBPs follows repeatable, cluster-dependent patterns. Anchoring curve modeling in historical on-chain data provides a robust foundation for simulation and optimization, improving both analytical rigor and practical relevance for future token launches.